

4(a)(i)	any <i>one</i> from: (membrane) fluidity will increase / AW ; increase in (lateral) movement of phospholipids (in bilayer) ; AVP : e.g. increase in passage of, polar molecules / ions, across membrane easier to fuse with other membranes	1
4(a)(ii)	idea that (greater membrane fluidity) makes it easier for lysosomes to fuse with, vacuoles / vesicles / phagosomes / endosomes / membranes (of other organelles) ; allow <i>ecf</i> on fluidity decreases in Q4(a)(i) e.g. makes more impermeable to prevent hydrolytic enzymes exiting	1
4(b)(i)	$K_m = 120$; unit = $\mu\text{mol dm}^{-3}$;	2
4(b)(ii)	any <i>four</i> from: differences 1 rate of reaction is lower at pH 4.5, throughout / at all substrate concentrations / AW ; ora 2 V_{max} / plateau, is 2.8 (at pH 4.5) v 5.9 (at 5.9) pmol min^{-1} ; 3 V_{max} is reached at a lower substrate concentration with pH 4.5 ; A ora A data 330–400 $\mu\text{mol dm}^{-3}$ compared with 700 $\mu\text{mol dm}^{-3}$ explanations to max 3 at pH 4.5 / lower pH 4 fewer enzyme-substrate complexes form ; 5 tertiary structure of enzyme / shape of active site, changes / AW ; 6 (at pH 4.5) active site shape becomes less complementary to substrate ; 7 detail ; e.g. increased presence of hydrogen ions has an effect on, R-group interactions / ability to lower activation energy / ability to bind substrate 8 suggestion that pH 5.9, is / is closer to, the optimum pH ; ora	4